Lab Work: Optical Thin Films Simulation 2026

From Transfer Matrix Theory to Cleanroom Vacuum Deposition at Institut Fresnel

FR FR | GB EN | Course portal | 🚀 Launch Web Simulator

10 MAJOR EXERCISES — 32 QUESTIONS — INDICATIVE DURATIONS

BUT 3 FI MCPC PATH — L2 · L3 · M1 · M2 STUDENTS

I II III IV V VI VII 🎯 VIII 🎯 IX X 🎯 XI 🔬 🎨 Free workshop

🎯 = cleanroom thread

🎬 Your mission

You are the optical engineer responsible for **designing, qualifying and then diagnosing** an interference filter that will be **physically deposited under vacuum** at Institut Fresnel. Between the first calculation and the spectrum measured on the finished component, you must prove — with numbers — that your design is not merely good on paper.

The simulator is not an oracle: it is a measurement bench. It will give you neither the full width at half maximum, nor the right merit function, nor the conclusion. You will enter your layers, take your readings with the cursor, export your data and conclude yourself. What counts is your reasoning: a number copied without explanation earns little, a sound but unfinished approach earns points. If you are stuck, open the 💡 **Hint** below the question: it puts you back on track without giving the answer.

⭐ Difficulty scale — every question is rated

★☆☆ Application

Reading, curve inspection, direct formula. If one of them blocks you, that is normal early in the lab: open its hint or reread the theoretical reminder, then carry on.

`7 questions — 22 %`

★★☆ Analysis

Several chained steps, data processing, physical interpretation. This is the core of the lab: allow time and use a scratch pad.

`15 questions — 47 %`

★★★ Expertise

You build the method, you decide and you argue. There is no single answer to copy: your reasoning is what is assessed.

`10 questions — 31 %`

> **How to use this:** the ★★★ questions are not "for the best students". They are the point of the module — the ★ and ★★ exist to get you there. If you are short of time, one ★★★ done thoroughly earns as much as three ★, and teaches you more. Work at your own pace: each question's hint is there for that.

🏷️ Indicative marking scheme

Each question is worth as many points as it has stars: ★ = 1, ★★ = 2, ★★★ = 3, i.e. 67 points for the 32 questions. The mark is scaled to 20 over the questions of your path only. A partial but reasoned answer earns points: always write down your approach, even if unfinished. The open workshop and exercise XI are not graded.

🧭 Suggested paths

L2: I, II, III and VI. **BUT 3 and L3**: I to VIII, then X. **M1 and M2**: everything, including IX, the open workshop and exercise XI. Your teacher may adapt this path.

📦 What you hand in

Session 1 (I–III): answers Q1–Q12 and the Q12 workbook tab. **Session 2** (IV–V): Q14 tab, tables of Q15–Q16, Q17 log. **Session 3** (VI–VII, start of VIII): tables of Q19–Q22, CSV and image of the Q23 spectrum, Q24 calculation. **Session 4** (VIII to X): Q25–Q26 with the yield plot, decision sheet for IX, diagnosis Q31–Q32. At every session, attach the .json file from "⬇ Export my work".

1. Pedagogical objective

This modular path covers the design, optimization, **manufacturability**, and reverse engineering of optical thin-film coatings (AR coatings, Bragg mirrors and resonant filters). A nominal minimum is never accepted as a completed design until process robustness has been demonstrated. Displayed durations are working estimates, not imposed limits; instructors may compose their own path.

> 🎯 **Cleanroom deposition:** only the **interference bandpass filter** designed in Exercises VII and VIII, then diagnosed in X, will be physically deposited under vacuum at **Institut Fresnel**, with the cleanroom deposition team.

🚀 Web Simulator Quickstart

- **Direct access:** Open simulateur_couches_minces.html in a browser. Network access is required to load the graphical libraries (an offline distribution exists: ask your instructor).
- **Virtual Spectrophotometer:** Build your stacks manually, take pointer measurements, and analyze data in Excel.
- **Symbolic formulas:** Fast formula entry such as `(HL)^3 2H (LH)^3`.
- **Exports:** Use **CSV** (for spreadsheet processing) and **Image** for your lab report.
- **Cursor reading:** hover over the chart; for a precise value, click it then step with ← → (Shift: ten steps). The readout box follows the displayed polarisation. The current merit is shown below the merit selector; the **× Scale** button multiplies thicknesses.
- **Published grids:** below the merit selector, the simulator displays **the exact grid** it uses. Reproduce it in Excel: you must recover the same number.
- **Universal CSV export: right-click any chart** (or long-press on mobile) to export all its visible curves on the grid actually plotted — including Monte-Carlo envelopes and saved curves.
- **Keyboard shortcuts:** press `?` for the full list. Most useful: `O` optimize, `N` needle, and `M` memorize.
- **Student welcome — choose your exercise:** click the current exercise card when the simulator opens, or choose it from the **Exercise** menu. The relevant panels and an instruction appear; the exercise merit (`Mean R` in IV and V, `Maximize Qr` in VIII and X) becomes available. The 🎛 **Free** button opens personal component design with your own target.
- **Free design:** press `D` or click 🎛 **Free** — you write *your own* specification. See the open workshop at the end of this document.
- **Active workbook:** use the Q12/Q14 scaffold; calculation cells are intentionally left for you to program.

🛡️ Mandatory protocol: from nominal optimum to a manufacturable, robust design

Decision rule: a stack that is optically excellent at nominal values but impossible to deposit, too thin, too complex, or too sensitive is **rejected**. Every industrial conclusion must separate nominal performance, manufacturability, and statistical yield.

1. **Specify before optimizing:** window, incidence, polarization, quantity, and acceptance threshold.
2. **Audit the nominal stack:** number of layers and materials, d_{min}, d_{max}, total thickness, and deposition order.
3. **Declare the process model:** Δd, minimum thickness, and the layers actually perturbed.
4. **Declare the acceptance tolerances:** without them the yield is undefined — it is the fraction of draws that stay inside the specification.
5. **Make the study reproducible:** record the random seed; compare two designs using the same seed.

5. **Size the sample:** $N = 100$ for exploration, $N = 500$ for comparison, and $N = 1000$ for educational qualification.
6. **Report robust statistics:** yield, 95% Wilson CI, mean, sample standard deviation, and P05/P50/P95.

7. **Use the adverse percentile:** P95 for a cost to be minimized; P05 for a performance to be maximized.

8. **Decide:** *qualified*, *inconclusive* (study to be extended), or *not qualified*, with margins and assumptions.

Warning: even 1000 successes out of 1000 do not prove a true 100% yield. The displayed confidence interval captures sampling uncertainty; it does not cover index uncertainty, thermal drift, roughness, or unmodelled defects.

Classic trap: the displayed P05/P50/P95 percentiles cover only the *manufacturable* realizations. They are therefore **conditional**. The yield, by contrast, is reported over the total number of runs. Never compare the two without saying so.

Theoretical Background: Interference & Transfer Matrix Method

💡 **Physical Principle:** Coatings rely on **multiple beam interference**. Optical path differences govern reflection and transmission. The core metric is the **Optical Thickness** ($n \times d$).

For each layer i with thickness d_i and complex index $\hat{n}_i = n_i - i\kappa_i$, the optical phase shift is:

$$\delta_i = \frac{2\pi d_i}{\lambda}\sqrt{\hat{n}_i^2 - n_0^2 \sin^2\theta_0}$$

The characteristic transfer matrix of layer i, which is unimodular ($\det M_i = 1$), is:

$$M_i = \begin{pmatrix} \cos\delta_i & \frac{i}{\eta_i}\sin\delta_i \\ i\eta_i \sin\delta_i & \cos\delta_i \end{pmatrix}$$

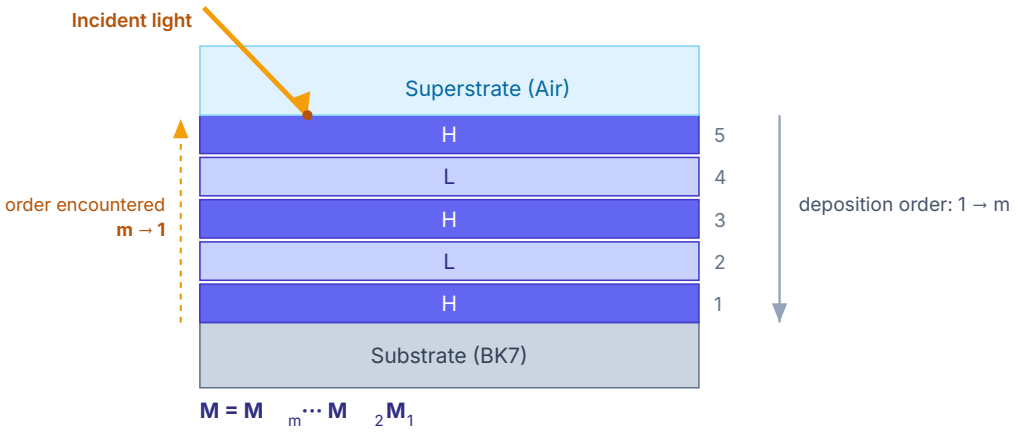
Polarization States (Incidence $\theta > 0$):

- **S-Polarization (TE):** $\vec{E} \perp$ plane of incidence $\implies \eta_i = \gamma_i = \sqrt{\hat{n}_i^2 - n_0^2 \sin^2\theta_0}$.
- **P-Polarization (TM):** $\vec{E} \parallel$ plane of incidence $\implies \eta_i = \hat{n}_i^2/\gamma_i$.
- **Unpolarized light:** $R_{avg} = \frac{R_s+R_p}{2}$ and $T_{avg} = \frac{T_s+T_p}{2}$.

📐 **Ordering convention — applies throughout this lab:**

Layers are listed in **deposition order, substrate → superstrate**. Layer 1 touches the substrate; the last one touches the air. Light therefore encounters them in reverse order, and the total matrix is $M = M_m \cdots M_2 M_1$.

Ordering convention: substrate → superstrate



The layer table follows **deposition** order: layer 1 touches the substrate. Light, however, arrives from the air and meets the layers in **reverse** order. Hence the matrix product $M = M_m \cdots M_1$. Getting the order wrong reverses the whole stack: it is the most common mistake in this lab.

I. Thin-film materials & complex index

⏱ 35 min indicative

Direct entry — prerequisites: no previous exercise. Be able to read a graph and handle a complex number written as $\hat{n} = n - i\kappa$.

Q1 ★☆☆

In the bottom *Materials database* panel, inspect dispersion curves $n(\lambda)$ and $k(\lambda)$ for **Si, SiO2, Ge, Al and Au**. Explain the physical nature of each material. Which parameter (n or k) dictates opacity?

💡 **Hint**

Look at k first: a transparent material has k much smaller than 1. Then compare the order of magnitude of k between families: dielectrics, semiconductors, metals. Course §2.2.

Q2 ★★☆

Rank the following six materials by increasing k: **SiO2, YF3, ZnS, Si, Al, Au**. Do it first **at 550 nm**, then rank them again **at 400 nm**: which materials change places? Finally, state which ones satisfy $k < 5 \times 10^{-4}$ *at every point* of the visible range $(400 - 750\,\mathrm{nm})$.

> **Careful — this is the heart of the question:** ZnS is transparent in the green and red but *not* in the deep blue. Record its k at 400 nm and then at 550 nm. What happens if you inspect a single wavelength only? Also discuss what a k displayed as **exactly** zero means: is it a measurement or a modelling convention?

💡 **Hint**

> Hover over the k(λ) curve to read values and build a small material × wavelength table before ranking. For the threshold, look for the largest k over 400–750 nm, not its value at 550 nm.

Q3 ★★☆

Compare orders of magnitude of k between metals (Al, Au) and semiconductors (Si, Ge). Justify with free carriers and bandgap E_g.

💡 **Hint**

> What absorbs light: free electrons (metal) or transitions above the gap (semiconductor)? Convert λ to energy: E (eV) ≈ 1240 / λ (nm).

Q4 ★☆☆

Examine **ZnS (High index H)** and **YF3 (Low index L)** used at the Institut Fresnel cleanroom. Record their indices at $550\,\mathrm{nm}$ and compute the contrast $(n_H - n_L)/(n_H + n_L)$.

> 🔑 **Keep this number:** it sets the mirror stopband width, $\Delta\lambda \simeq \frac{4\lambda_0}{\pi}\arcsin\frac{n_H - n_L}{n_H + n_L}$, which you will need from Exercise VI onwards and above all in VIII (Q24) to assess where the measurement bands are placed. Beware: this width is symmetric in wavenumber, not in wavelength — the edges lie at $\lambda_0/(1 \pm \frac{2}{\pi}\arcsin\frac{n_H - n_L}{n_H + n_L})$, so the band extends further on the red side than on the blue side.

💡 **Hint**

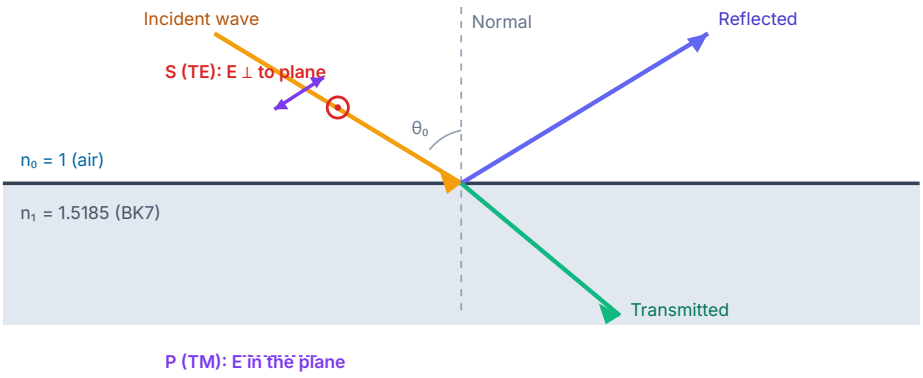
> Read n at 550 nm for ZnS and YF3 in the materials database, then apply the formula: it is a simple ratio.

II. Single interface: BK7 // Air

⏱ 50 min indicative

> **Direct entry — prerequisites:** Fresnel coefficients at one interface, incidence-angle definition and S/P distinction. Exercise I is recommended only for dispersion and complex index.

S and P polarizations at oblique incidence



In S (TE) the electric field is **perpendicular to the plane of incidence** (⊙ symbol). In P (TM) it **lies in that plane** (double arrow). The two see different admittances, $\eta_s = \gamma$ and $\eta_p = \hat{n}^2/\gamma$: this is the whole origin of the Brewster angle, where R_p vanishes while R_s never does.

Q5 ★☆☆

Hand Calculation: Compute normal incidence R and T for the Air/BK7 interface ($n_{BK7} \approx 1.5185$ at $550\,\text{nm}$) with Fresnel equations.

> 🔍 **Self-check:** your R must fall between 3% and 5%. If you get 0.04, you forgot to convert to a percentage; if you get 20%, you confused r with $R = |r|^2$.

💡 **Hint**

> At normal incidence, r = (n₀ − n₁)/(n₀ + n₁) and R = |r|². Without absorption, T = 1 − R.

Q6 ★★☆

Set Superstrate = `Air` and Substrate = `BK7` without layers. Compare the simulator value at $550\,\text{nm}$ with your hand calculation. Then tick **Include rear surface**: reflection rises from about 4.2% to 8.1%. Recover this second figure from the incoherent sum $R_{tot} = R + \frac{(1-R)^2 R}{1-R^2}$, then show that it reduces to $R_{tot} = \dfrac{2R}{1+R}$ — a shorter and strictly equivalent form. Finally, explain why the result is *not* simply $2R$.

The formula above assumes two **bare, identical** interfaces: it no longer holds once one face is coated, even though the simulator can then compute the whole component. **Untick the box before going on**: every following question deals with the coated face alone, on a semi-infinite substrate.

💡 **Hint**

> Follow one ray: it reflects on the front face, or goes through and bounces back and forth in the glass. Add the intensities: it is a geometric series of ratio R².

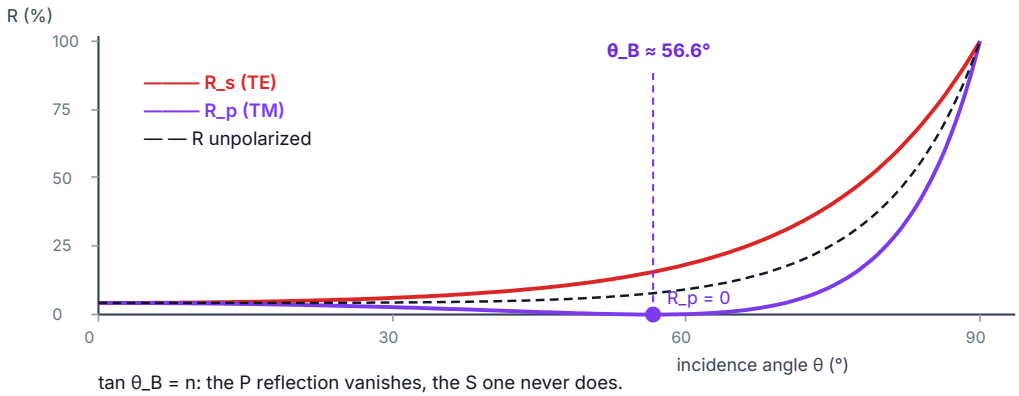
Q7 ★☆☆

Switch to `Angle θ` mode (at $\lambda = 550\,\text{nm}$). Record $R(\theta)$ for $\theta \in [0, 80°]$ in $10°$ steps under unpolarized light, then separately in S and in P. Plot all three curves.

Hint

> Angle θ mode, then choose S, P or the average. Hover over the chart, or click it and step with ← →: the readout box gives the value. The CSV export also lets you read the values every 10°.

Reflectance versus angle: S, P and Brewster



tan θ_B = n: the P reflection vanishes, the S one never does.
Unpolarized, R only drops to half of R_s: the minimum is not zero.

Record these three curves yourself in `Angle θ` mode. The remarkable point is the vanishing of R_p at θ_B = arctan(n). Note that the unpolarized curve, the average of the two, **never vanishes**: this is why polarizing sunglasses remove a window's glare while a neutral filter cannot.

Q8 ★☆☆ **Brewster Angle:** Find angle θ_B where P-polarized reflection vanishes ($R_p = 0$). Verify $\tan\theta_B = n_{BK7}$. Cite a real-world application.

Hint

> Display only the P polarisation: the readout box then gives Rp. Find where it touches zero and zoom around that point. Then compare arctan(n) with the angle you read.

Q9 ★☆☆ **Total Internal Reflection:** Set Superstrate = `BK7` and Substrate = `Air`. Identify θ_{lim} and link it to Snell-Descartes law and optical fiber waveguides.

Hint

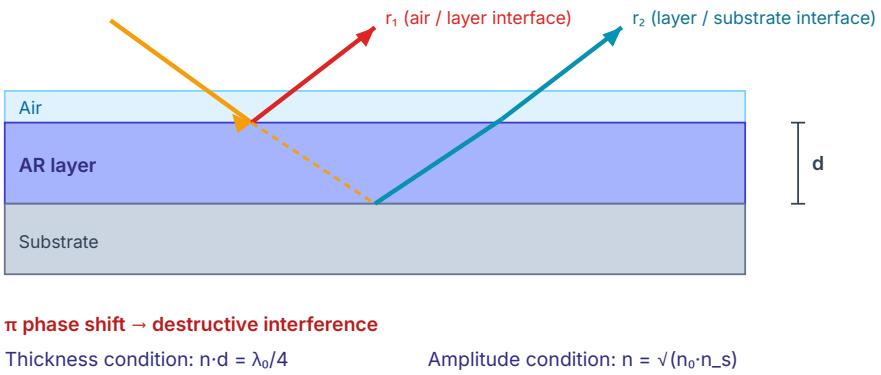
> After swapping the media, find the angle beyond which R equals 100%. Snell's law: $n_1 \sin\theta_1 = n_2 \sin\theta_2$, with $\theta_2 = 90°$.

III. Single-layer antireflection coating

⏱ 60 min indicative

Direct entry — prerequisites: single-interface reflection, optical phase $2\pi nd/\lambda$ and elementary complex matrix calculation. Complete at least Q5 in Exercise II if Fresnel coefficients are not yet secure.

Quarter-wave antireflection: the two conditions



Ray r_2 travels **2·n·d** further than r_1. For $n \cdot d = \lambda_0/4$ this difference is $\lambda_0/2$: the two waves are in antiphase. But cancellation is **complete** only if their amplitudes match, which requires $n = \sqrt{n_0 \cdot n_s}$. Two independent conditions — question Q10 has you verify that the second one is not exactly met here.

💡 **AR principle:** quarter-wave condition $n \cdot d = \lambda_0/4$ with $n_{layer} = \sqrt{n_{substrate} \cdot n_{inc}}$. **QWOT** (*quarter-wave optical thickness*) is the optical thickness $\lambda_0/4$: a 1 QWOT layer satisfies $n \cdot d = \lambda_0/4$, and "2H" means 2 QWOT of H.

Q10 ★★☆

Manually add the theoretical material `Ideal AR (n ≈ 1.225)` with $d = 112.3\,\text{nm}$. Find the spectral window where $R < 0.5\%$. Prove that $112.3\,\text{nm}$ is indeed the quarter-wave thickness at $550\,\text{nm}$ for this index.

🧠 **Deliberate trap, to be exposed:** the supplied index is exactly $1.22474 = \sqrt{1.50}$. But real BK7 has $n = 1.5185$ at 550 nm, so the *ideal* index would be $\sqrt{1.5185}$. Compute that ideal index, the corresponding quarter-wave thickness, and then the residual R actually obtained at 550 nm with the supplied material. Is it rigorously zero? Conclude: under what condition may one write "perfect antireflection"?

💡 **Hint**

A quarter-wave layer satisfies n·d = λ₀/4: compute d = λ₀/(4n). For the R < 0.5% range, read both edges with the cursor (click the chart, then ← → to move one step).

Q11 ★★☆

Does substrate dispersion actually matter? First record n_{BK7} at 400, 550 and 750 nm: the variation exceeds 1%. One therefore expects an effect.

- Carefully note the position of the R minimum and the bounds of the $R < 0.5\%$ range with the `BK7` substrate.
- Replace the substrate with `Custom index` at a **constant** $n = 1.5185$, and repeat both measurements.

🧠 **The difference is below one nanometre — hence negligible.** This is not an error: it is the result. Explain it. Hint: the *position* of the minimum is set by the phase condition $n_c d = \lambda/4$, which involves only the layer — non-dispersive here. Substrate dispersion acts solely on the *amplitude* condition $n_c = \sqrt{n_s}$, hence on the **depth** of the minimum, not on its wavelength.

Conclude by naming, within this lab, **two situations where dispersion does become visible**: you have met one already in Exercise I, and you will meet the other in Exercise VI.

💡 **Hint**

Record the position of the minimum before and after changing the substrate, with the cursor and the same step. The hint at the end of the question tells you what you should find.

Q12 ★★★

Demystifying the Black Box in Excel: Compute reflection R of this layer at $550\,\mathrm{nm}$ in Excel.

- Compute the optical phase δ_1.
- Build matrix M_1 using `=COMPLEX(real, imag)` and `=IMPRODUCT(z1, z2)`.
- Compute the equivalent admittance $Y = \frac{m_{10}+\eta_{sub} m_{11}}{m_{00}+\eta_{sub} m_{01}}$ then $R = \left|\frac{\eta_0 - Y}{\eta_0 + Y}\right|^2$.
- Check that you recover the simulator value.

🔍 **Self-check:** your R must match the simulator to better than 0.001 point. If you get a negative R or one above 1, the error is in the squared modulus. If the gap is of order one percent, check the sign of i in the off-diagonal terms of M_1.

💡 **Hint**

Go cell by cell and check each number: δ₁ first (a quarter-wave layer gives π/2), then the four matrix terms, then Y. Course §4.1 and §12.

IV. Double-layer antireflection: BK7 / H / L / Air

⏱ 85 min indicative

Direct entry — prerequisites: QWOT, substrate-to-superstrate order, matrix products and reflectance reading. Exercise III, especially Q10–Q12, is strongly recommended.

$H = \mathrm{ZnS}$ (high index), $L = \mathrm{YF_3}$ (low index).

Q13 ★☆☆

Add two layers (ZnS then YF_3). Manually adjust thicknesses d_{ZnS} and d_{YF3} ($< 150\,\mathrm{nm}$) to minimize R at $550\,\mathrm{nm}$. Stop as soon as $R < 0.5\%$, or after ten changes that bring no gain. **Count one change each time you edit a thickness**, and record that total: it measures the cost of a search by hand, over only two parameters.

💡 **Hint**

Start from two quarter-wave layers, then change only one thickness at a time, in 10 nm steps; when R rises again, step back and reduce the step.

Q14 ★★☆

2D Mapping in Excel: Explore the parameter landscape topology. At $\lambda = 550\,\mathrm{nm}$:

- Record R on a grid: d_{ZnS} from **20 to 140 nm** and d_{YF3} from **20 to 140 nm**, in **30 nm** steps — 25 readings, to be entered in the `Q14_Cartographie` sheet of the workbook. *These bounds are not arbitrary: they span roughly one oscillation period of R in d_{ZnS} ($\lambda/2n_H \approx 116$ nm). A coarser step would under-sample the landscape and the valleys would vanish.*
- Enter the table into Excel and create a **3D Surface or Contour Plot**.
- Locate the coordinates of the minimum valley.
- Is your Q13 minimum inside one of these valleys? How many distinct valleys do you see, and what does the 30 nm step tell you about the exact position of their floors?

💡 **Hint**

Fill the grid row by row, changing only one thickness at a time. In Excel, the Surface or Contour chart is built from the table and its two headers.

Q15 First robust qualification — but over which window?

★★★

🧠 **Specification trap, to be dealt with first.** In 🎓 **Student** mode, exercise IV, the `Mean R` merit is offered by default over $400 - 750\,\mathrm{nm}$. But you have just designed an antireflection coating *at 550 nm*. Read the merit value over that wide window and compare it with bare glass: depending on the valley you kept, it can be much worse. **Is your design bad, or is your specification off-topic?** Justify, then correct the window.

- First refine your Q13 design down to the bottom of its valley ($R < 0.01\%$ at 550 nm) and note which Q14 minimum you kept: everything that follows depends on it.
- Set the **AR window to 520–580 nm** and the **threshold to 0.8%** (fields below the merit selector). Check that the nominal design now meets the criterion, and with what margin.
- Qualify: $d_{min} = 1$ nm, $N = 500$. Compare $\Delta d = 3$ and $5\,\mathrm{nm}$; the seed being fixed, both runs use the same realizations.
- Report yield, the 95% Wilson CI and P95 of mean reflectance in both cases. Do the two confidence intervals overlap? Can you claim ± 5 is significantly worse, or is it sampling noise?
- Compare P95 at $\pm 5\,\mathrm{nm}$ with the threshold. Depending on your valley, it stays below or exceeds it while the mean respects it: in the latter case, what does that gap mean concretely for a production batch? Compare with a pair who kept another valley: which one is more robust, and why?
- **Isolate the critical layer — by hand, not with Monte-Carlo.** The draw *always* perturbs both layers: locking applies to BFGS only, never to manufacturing. So work by sensitivity instead: from the nominal design, add $+3\,\mathrm{nm}$ to ZnS alone and record the merit; return to nominal, add $+3\,\mathrm{nm}$ to YF_3 alone and record. Which one degrades it more? Relate your answer to the index and the optical thickness $n \cdot d$ of each layer.

🔍 **Self-check:** with the right window, the yield at $\Delta d = 5$ nm must be **lower than or equal to** the one at $\Delta d = 3$ nm — that is the expected ordering, even though it is not mathematically guaranteed. If you get 0% everywhere, check that the window is 520–580 nm and that your nominal design meets the threshold: if it does not, the yield may even grow with Δd — explain why. If you get 100% in both cases, your nominal margin is too comfortable for the test to discriminate: tighten the threshold (try 0.3% instead of 0.8%) and rerun both. A test that separates nothing does not allow a conclusion: tightening it is how you learn something.

Keep the idea in mind: a standard deviation alone is not enough to conclude on robustness; you need an acceptance criterion.

💡 **Hint**

> Follow the bullets in order. The “Current merit” line, below the merit selector, updates with every change: read it before and after changing the window. The Monte-Carlo result directly gives yield, interval and P95. Course §11.

Q16

★★☆

Statistical convergence. Since the seed is fixed, the draws are strictly identical from one run to the next. Take your Q15 design in a setting where the yield is intermediate — for example the tightened threshold of 0.3% and $\Delta d = 5$ nm — and run $N = 100$, 500 then 1000. Tabulate the yield and the bounds of the 95% Wilson CI, and measure the interval width in all three cases: by what factor does it narrow when N is multiplied by 10? Does that match the $1/\sqrt{N}$ law? Repeat the test in a setting where the yield is close to 100% (threshold 0.8%, $\Delta d = 3$ nm): why does the law no longer apply? Conclude: why are 100 runs enough to explore, but not to qualify?

💡 **Hint**

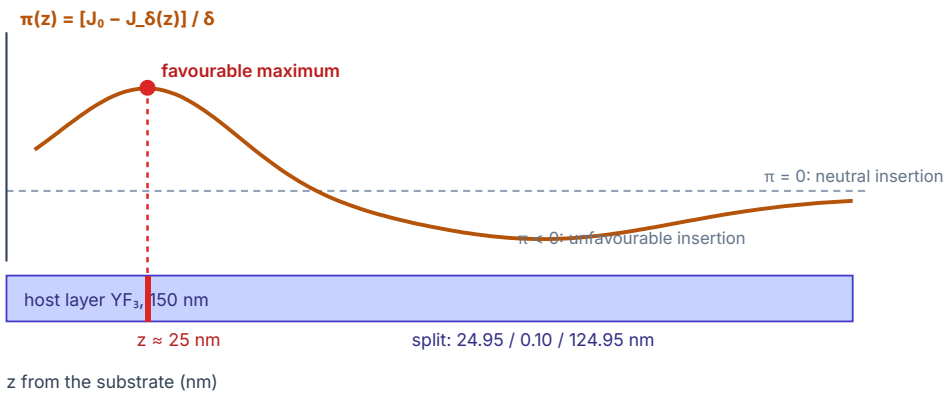
> Change only N from one run to the next. Interval width = upper bound − lower bound; divide the width obtained at N = 100 by the one obtained at N = 1000.

V. Multilayer synthesis & the Needle method

⏱ 90 min indicative

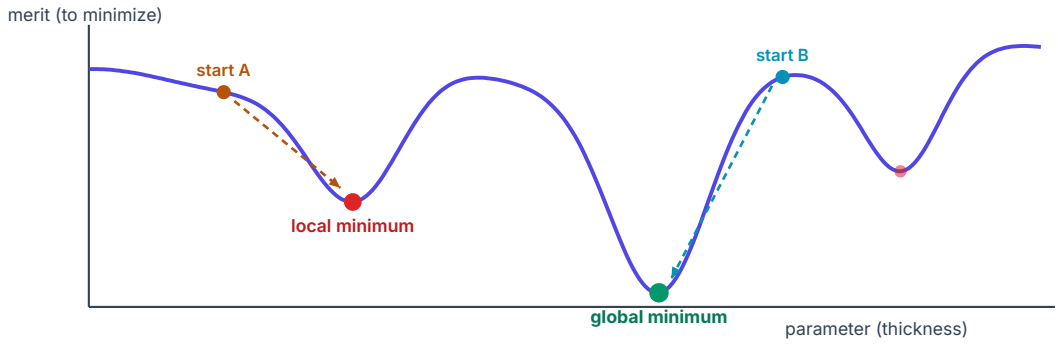
> **Direct entry — prerequisites:** be able to define a merit function, distinguish global exploration from local refinement, and build a simple AR coating. Exercises III and IV are recommended; Q14 may be skipped if local minima are already understood.

The needle diagnostic: read π(z) before inserting



π(z) measures the merit gain an infinitely thin needle would bring at depth z. Where π > 0, inserting improves; where π < 0, inserting degrades. You insert at the **maximum**, then refine with BFGS. Careful: the scan is discrete (5 nm) and never evaluates the interfaces — Q18 asks you to work out the consequences.

Why the starting point decides the result



BFGS is deterministic: same start, same result. Different start, different minimum.

Optimizing only goes downhill. To change valley you must change the problem: that is what the needle does by creating a la

The merit function has more than one valley. **BFGS descends** into the nearest one and stops: two different starting stacks therefore give two different results, and neither is "the optimum". Since the algorithm is deterministic, re-running changes nothing — what must change is the **starting point**, or the **structure** of the stack. Adding a layer with the needle method adds a dimension to the landscape: a barrier that cannot be crossed in dimension *n* can be bypassed in dimension *n+1*.

Q17 ★★☆

BFGS local-convergence study. In 🎓 **Student** mode, exercise V, activate the `Mean R` merit over **400–750 nm** with a **0.5%** threshold. Goal: a *broadband* antireflection coating on BK7 using ZnS/YF_3.

🧠 **Start with the instructive mistake.** First record the merit of **bare glass** (no layer): that is your baseline. Then build the most natural stack — quarter-wave layers `H L`, then `H L H L`. **The merit is WORSE than bare glass — clearly for `H L`, massively for `H L H L`.** This is not a handling error. Explain it: from an interference standpoint, what does a quarter-wave *high*-index layer on glass actually do? Must an antireflection coating be a quarter-wave stack?

- Restart from **thin, asymmetric layers** (a few tens of nanometres), not quarter-waves. The optimizer is a **local descent**: it reaches the minimum of the valley you drop it into, and looks for no other.
- Run it from **at least three different initial stacks** and record, each time, the initial merit, the final merit and the layer count after cleanup. Since the algorithm is deterministic, the same start returns exactly the same result: every bit of spread you observe comes from the **starting point**, not from the algorithm.
- **How many layers are needed?** Find the best result you obtain with 2, then 3, then 4 layers. For each case, decide whether the 0.5% threshold is crossed and quantify the margin.
- A *starting point* that stays stuck far above the others tells you nothing about the optimizer — there is only one, and it is deterministic. It tells you about *that starting point*. Verify this by rerunning from a different initial stack.

> The exact merit grid is displayed below the selector. Copy it into your report: without it, your value is not reproducible by a third party.

💡 **Hint**

> For bare glass, empty the stack and read the "Current merit" line. As a thin, asymmetric start, try for example 20 nm and 40 nm. Record everything in a table: start, initial merit, final merit, number of layers. Course §10.

Q18

★★★

Guided Synthesis via the Needle Method:

- Start with a single layer: YF3 with physical thickness $150\,\mathrm{nm}$ on BK7 glass.
- Click **Analyze the needle**. Read the diagnostic carefully: what material and depth z does the perturbation analysis recommend?
- The diagnostic numerically plots $\pi(z) = [J_0 - J_\delta(z)]/\delta$ every $5\,\mathrm{nm}$ using a virtual needle $\delta = 0.1\,\mathrm{nm}$. Locate its positive maximum and the position z from the substrate, then insert the recommended needle.
- Click ⚡ **Insert + optimize with BFGS**: the software creates the virtual $0.1\,\mathrm{nm}$ layer and then automatically runs BFGS. Compare the merit immediately before insertion and after this refinement.
- **Critical analysis:** explain the value of the two-step workflow (nucleation diagnostic → local refinement).

🧠 **Three limitations to identify yourself** (the π(z) tooltip mentions two: it is up to you to analyse them): (a) the scan is *discrete*, every 5 nm — what if the optimum lay 2.5 nm from a tested point? (b) $\pi(z)$ is **never** evaluated at the interfaces, although that is often where the true optimum sits: why this choice, and what is missed? (c) the derivative is approximated by finite differences with a finite δ. Quantify at least one of these three limitations with a numerical test.

💡 Hint

The π(z) curve appears after "Analyze the needle"; its positive maximum is the recommended place. Record the merit just before clicking "⚡ Insert + optimize with BFGS". Course §10.

VI. Bragg mirrors: BK7 / (HL)^n H / Air

⏱ 55 min indicative

Direct entry — prerequisites: QWOT, constructive interference and spectrum reading. Exercise III is sufficient; Exercises IV and V are not required.

💡 **Bragg mirror principle: constructive** interference from alternating quarter-wave layers ($n \cdot d = \lambda_0/4$).

Q19 ★★☆

Enter formula `(HL)^3 H` at $\lambda_0 = 550\,\text{nm}$. Measure peak reflectance R_{max} and record **both edges** of the **stopband** — the wavelength range over which the mirror reflects strongly, also called the photonic band gap — using $R \geq 90\%$ as the convention. Note the width you obtain.

> 🧠 **Do not rush to the formula.** Compare your measured width with the prediction $\Delta\lambda = \frac{4\lambda_0}{\pi}\arcsin\frac{n_H - n_L}{n_H + n_L}$ computed in Q4: **the gap is large, of order one third.** This is not a measurement error. Look at R_{max}: by how much does it exceed your 90% threshold? What happens to the measured width if R_{max} barely clears the threshold? Formulate a hypothesis — you will test it in Q20.

> 🔑 **Result to keep: you will use the stopband edges in VIII (Q24)** to assess where the measurement bands are placed: a rejection band placed where the stack no longer blocks does not measure the filter, it measures a leak. You will refine them in Q20 on a thicker mirror.

💡 **Hint**

> For the edges, find with the cursor the two wavelengths where R crosses 90% again. The Q4 formula is explained in course §9.

Q20 ★★☆

Compare `(HL)^3 H` (7 layers), `(HL)^4 H` (9), `(HL)^5 H` (11) and `(HL)^6 H` (13) using **Save curve** overlays. For each, record R_{max} **and the width at** $R \geq 90\%$. Build the table of the four pairs.

> 🧠 **The central result of this exercise.** Both quantities saturate, but not in the same way. Now confront your four widths with the Q4 prediction: it describes *none* of your four mirrors exactly, yet it describes very well the one they converge towards. **So what does that formula physically represent** — the width of a real mirror, or a limit? Is your Q19 hypothesis confirmed?

Conclude: from how many periods onwards can the formula be used as an acceptable approximation? That is the value you will use in Q24 to place your measurement bands.

💡 **Hint**

> Click “Save curve” after each stack to overlay them, and fill in a table: periods, R max, width.

Q21
★★☆

The outermost layers: a setting, not an improvement. On the 13-layer mirror, widen the display to $360 - 800\,\mathrm{nm}$ so you can see the ripples on both sides of the stopband.

- **Reference:** record three numbers — R_{max} inside the band, the **highest sidelobe on the blue side** $(360 - 450\,\mathrm{nm})$ and the **highest sidelobe on the red side** $(660 - 800\,\mathrm{nm})$.
- Set layers **1 and 13 to** $0.5\,\mathrm{QWOT}$ and repeat the three measurements.
- Now set them to $1.3\,\mathrm{QWOT}$ and repeat again.

🧠 **What you will discover, and it is not what one expects.** The ripples are not *suppressed*: they are **transferred from one side of the stopband to the other**. Thinning the outermost layers cleans one side and degrades the other; thickening them does the opposite. Explain this in terms of **admittance matching** — the admittance η from the theory reminders: what does the wave see on arriving at the first interface, depending on the thickness of the outermost layer?

Finally look at R_{max}: does it change much? Conclude on the nature of this setting — is it a performance gain, or a **trade-off** the designer chooses according to which side of the spectrum must stay clean? Give an example of a component where this choice is forced.

🔍 **Self-check:** if both sidelobes move in the same direction, you have measured only one side of the band, or your display range is too narrow.

💡 Hint

Change only layers 1 and 13 in the layer table (thickness in QWOT). Always record the same three numbers before comparing.

VII. Interference bandpass filter (Fabry–Pérot)

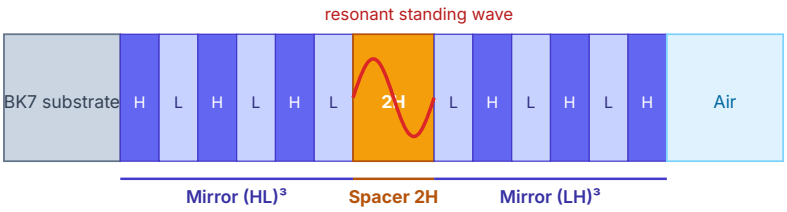
⏱ 75 min indicative

🎯 Deposited at Fresnel

Direct entry — prerequisites: Bragg mirror, resonant cavity and manual spectral-width measurement. Complete Exercise VI or review the Bragg reminder first; optimizer knowledge is not required.

💡 **Fabry–Pérot cavity:** two Bragg mirrors separated by a central half-wave spacer ($2H$ or $2L$). The more reflective the mirrors, the higher the finesse and the narrower the peak.

Anatomy of the filter: two mirrors and a spacer



(HL)ⁿ 2H (LH)ⁿ

n ↑: more reflective mirrors → narrower peak

q ↑: thicker cavity → narrower peak, but spurious orders

The half-wave **2H** spacer is a cavity: at resonance a standing wave builds up inside it and light passes almost entirely through the stack, while it is blocked everywhere else in the stopband. Two independent levers narrow the peak — increasing **n** (more reflective mirrors) or increasing **q** (thicker cavity). Q22 asks you to determine which acts more strongly.

Q22

★★☆

At $\lambda_0 = 550\,\text{nm}$, enter `(HL)^2 2H (LH)^2`, then `(HL)^3 2H (LH)^3`, then `(HL)^4 2H (LH)^4`. Measure the full width at half maximum of each.

Then test the cavity order: replace the `2H` spacer with `4H` (order $q = 2$) then `6H` ($q = 3$). *Notation: the prefix is the QWOT multiple, so `4H` = 4 quarter-waves = 2 half-waves.*

Establish the relationship between the number of mirror periods n, the cavity order q, and the resolving power $\lambda/\Delta\lambda$. Which of the two parameters acts most strongly?

💡 **Hint**

> For each filter: T max, then T max / 2, then the two wavelengths where T equals T max / 2. Box zoom (drag on the chart) helps a lot. Course §9.

Q23 ★★☆

Fresnel prescription that will actually be deposited: load `BK7 / H L 2H L H L H L 2H L H L H L 2H L H L / Air` at $\lambda_0 = 550\,\mathrm{nm}$ (**Symbolic formula** button, then **Load the Fresnel reference**). Verify yourself that it contains exactly **18 layers**, in substrate-to-air order. This architecture is mandatory: it must neither be shortened nor replaced by a symmetric cavity.

Mandatory manual measurement. With a $0.1\,\mathrm{nm}$ spectral step, read from the cursor: (a) the maximum T_{max} and its wavelength, (b) the $T_{max}/2$ level, (c) the **two** intersections λ_1 and λ_2, (d) the width $\lambda_2 - \lambda_1$, (e) the **half-maximum centre** $(\lambda_1 + \lambda_2)/2$. Use box zoom to refine each intersection, then `Reset view` to return to the full spectrum.

This prescription contains **three $2H$ spacers**, hence three coupled cavities: that is what makes it a *multicavity* filter. Its passband is far wider than the single-cavity filters of Q22 — do not transfer their expected scale — and above all its top is **flat and rippled**. You will find several near-equal maxima there, and the highest one can jump from one to another after a tiny change in the stack. **So it is the half-maximum centre (e), not the position of the maximum, that locates the passband reliably**; that is the figure you will reuse in Exercises IX and X.

> 🔍 **Method self-check:** measure at half the *window* height ($T_{max}/2$), never at 50% of the scale. This filter has three cavities: its top is flat and rippled — it is not a peak. The simulator's display step is fixed at 0.1 nm: to quantify the effect of the step, export the CSV and, in Excel, keep only one row in 5 (0.5 nm step) and then one row in 20 (2 nm step). Read T_{max}, λ_1 and λ_2 again on each grid and quantify the reading uncertainty caused by the step.

> 🎯 **Deliverable to keep:** export the nominal CSV and the spectrum image. **This same 18-layer stack is reused without architectural change in Exercises VIII to X, then manufactured at Fresnel.** Without this file, the rest of the project cannot be completed with full traceability.

💡 **Hint**

> "Symbolic formula" button, then "Load the Fresnel reference". Measure at half of T max, not at 50%. For the step effect, the CSV export and a row filter in Excel are enough.

VIII. Cleanroom pre-fabrication project

⏱ 100 min indicative

🎯 Deposited at Fresnel

> **Direct entry — prerequisites:** retain the nominal Q23 file and understand a cavity passband, a merit function and parameter locking. Exercises VI and VII are required; V is useful for Q25.

📋 Component specification

- Measured passband: **535 – 565 nm**
- Rejection band 1: **460 – 500 nm**
- Rejection band 2: **620 – 680 nm**
- Calculation grid: **0.5 nm step**

- Required rejection factor: $Q_r \geq 30$
- Required passband transmission: $\overline{T}_{pass} \geq 85\%$

⚠ **Why these bands?** The passband frames the cavity resonance; the two rejection bands are placed where the complete 18-layer stack blocks. Yet they extend beyond the stopband of the simple mirror measured in Q19–Q20: check this, then verify on the Q23 spectrum that T stays low over their whole extent. This is an engineering decision, not a natural given: in Q24 you will have to show what happens if it is made at random.

$$Q_r = \frac{2\,\overline{T}_{535-565}}{\overline{T}_{460-500} + \overline{T}_{620-680}}$$

Q24 ★★★

Nominal audit and manual Q_r calculation: reopen the Q23 filter, in the same substrate-to-superstrate deposition order.

⚠ **The analyzer does not compute Q_r.** It only gives you **measurements**: the average of R, T, or A over a band you define, at a step you choose. It affects neither the optimizer nor Monte-Carlo. The rejection factor is a *derived* quantity: first build it yourself; the BFGS merit will then use it with fixed bands.

- In the analyzer, select only `T`, enter the three specification bands with a 0.5 nm step, and record the three $\overline{T}$ averages. The `R` and `A` choices support other measurements; they never assign a role in a merit.
- Compute Q_r yourself from those three numbers, by hand and then in Excel from the exported CSV. Both routes must agree.
- Check the effect of the step: repeat the same averages at 0.1 nm and then at 2 nm. Which of the three bands is the most sensitive to the step, and why?
- Complete the nominal audit — number of layers, number of materials, d_{min}, d_{max}, total thickness — **by computing it from the layer table**. The *"check my audit"* button is only there to verify your result **after** you have established it.

🧠 **The decisive experiment:** keep the 535–565 nm passband and 620–680 nm upper rejection band, then **recompute Q_r three times**, changing only the lower rejection band: `400–500`, then `430–500`, then the prescribed `460–500`. **By how much does Q_r vary?** Relate this to the actual spectrum. What is a rejection specification worth if it does not state its bands?
Note: the 400–500 nm band goes beyond the plotted range (430–700 nm); the software says so, but the average does cover the whole requested band. Widen λ min to see what the 400–430 nm part contains, and say how much it weighs in the average.

🔍 **Self-check:** your manual and Excel calculations must agree within rounding. Check that all three bands are fully covered, that the measured quantity is T, and that the denominator is the *mean* of both rejection values, not their sum.

💡 **Hint**

Add three bands in the analyzer, quantity T, 0.5 nm step. Then Q_r = 2 × T̄(pass) / (T̄(reject 1) + T̄(reject 2)): check the brackets.

Q25

★★★

Is there really anything to optimize?

The question is not rhetorical. Start by measuring Q_r and $\overline{T}_{pass}$ *before* any optimization and compare them with the two requirements of the specification. **If the deposited prescription already meets them, and by a wide margin, say so: that is a result, not a failure on your part.** You then have to answer a more interesting question: what is the point of optimizing a component that already passes? Look at the margin — by how much does the nominal exceed the threshold, and does that margin survive the deposition errors of Q26? That, rather than the nominal value, is where the three free layers earn their keep.

- Switch to 🎓 **Student** mode, exercise VIII: the `Maximize Qr` merit is selected automatically there. Its formula already uses exactly the 460–500, 535–565, and 620–680 nm bands; the band analyzer remains an independent measurement instrument.
- Set the cleanup threshold to **2 nm**: it is also BFGS's lower bound. Record this constraint in the design dossier; a numerical optimum below it is not admissible.
- Choose `Specific layers`, then allow **layers 16, 17, and 18 only**. Check in the table that layers 1 to 15 are fixed. This constraint belongs to the component that will be deposited and must never be relaxed.
- Run BFGS and record Q_r, $\overline{T}_{pass}$, and the three thicknesses before/after. Do both quantities move in the same direction? What happens to layer 18? Verify that the layer count, ordering, and first fifteen thicknesses are strictly unchanged.
- Test reproducibility by reloading the prescription and rerunning BFGS with identical settings. Explain why identical results are expected from a deterministic algorithm.
- Reject any nominal improvement that creates a layer below d_{min} or lowers passband transmission below its threshold: an isolated Q_r score is not a manufacturing dossier.

💡 **Hint**

> First check that the “Maximize Qr” merit is selected and that only layers 16–18 are free. Record Q_r and T̄pass before running BFGS, then after.

Q26 ★★★

Pre-manufacturing review — robust qualification without supplied results.

- **(a)** Keep the Q25 optimized prescription: 18 layers, the first fifteen unchanged, the final three produced by BFGS. Set $d_{min} = 1$ nm and write, before any simulation, the acceptance criteria for one draw — at least the two specification requirements, $Q_r \geq 30$ AND $\overline{T}_{pass} \geq 85\%$; any tightening must be justified — together with the **minimum required yield** of the batch, entered in the `Required yield` field: this is the value used in (d).
  Two different thresholds, not to be confused: the **2 nm** of Q25 is a lower bound on what *the optimizer* may propose; the $d_{min} = 1$ **nm** here is the thinnest layer one accepts to *deposit*, and it is used to sort the Monte-Carlo draws. The first constrains a calculation, the second judges a fabrication. Justify both values in your dossier.
- **(b)** First check the protocol with a single $N = 100$ run at $\Delta d = 2$ nm, then run $N = 1000$ for $\Delta d = 1$, 2 and 5 nm. Keep the seed unchanged.
- **(c)** For each severity, report yield, 95% Wilson CI, P05(Q_r), P05($\overline{T}_{pass}$), manufacturability rate, and number of perturbed layers. Monte-Carlo perturbs the **18 deposited layers**: locking the first fifteen applied to BFGS only.
- **(d)** Plot yield and both adverse percentiles against Δd. By cautious interpolation, determine the largest Δd compatible with the minimum required yield; if the data do not decide, give an interval and propose the additional run needed.
- **(e)** Clearly distinguish nominal performance, statistical robustness, and manufacturability. Base the deposition recommendation on confidence bounds, without treating Δd as an absolute bound: here $\sigma = \Delta d/2$.
- **(f)** Reload the nominal Q23 prescription and qualify it with exactly the same settings, at $\Delta d = 2$ then 5 nm, $N = 1000$. Which of the two prescriptions do you send to the cleanroom? Justify with the yields and their intervals, not with the nominal Q_r.

🔬 **Evidence requirement:** no yield curve or expected value is supplied. Your plot must show the 1, 2, and 5 nm points, 95% CI bars, seed, N, thresholds, and error model. Without a confidence interval, or with settings that change between runs, the conclusion cannot be checked: remember to record them.

💡 **Hint**

Set the criteria, d_min and the required yield once and for all, then change only Δd. The Monte-Carlo result gives yield, interval, P05 and manufacturability. Course §11.

IX. Design review — industrial qualification dossier

⏱ 75 min indicative

M1 · M2 — optional for BUT 3 / L3

Audience: exercise intended for M1/M2, optional for BUT 3 and L3 — the choice between nominal and optimized prescription is already made in Q26 (f). **Direct entry — prerequisites:** complete Q15 and at least one full qualification among Q26 or a personal extension; distinguish independent error, systematic bias, percentile, and confidence interval. The other filter families may be skipped.

Deliverable: a one-page decision sheet detailed enough for another team to reproduce the simulation and challenge the conclusion.

Q27 ★★☆

Qualification contract: work on the Q25 optimized Fresnel prescription (layers 16 to 18), or on a previously approved custom extension whose yield is not close to 100%. Switch the simulator to `Student → IX. Design review`. Check the external media, then state before any calculation the window, incidence, polarization, metric, **the exact bands**, calculation grid, optical threshold, d_{min}, resolution, seed, and minimum required yield. Enter the latter in the *Required yield* field. Distinguish requirements from improvement goals.

💡 Hint

> This is about writing, not computing: list every setting that changes the result, with its value.

Q28 ★★☆

Nominal manufacturability audit: produce the material/thickness/order table, then **establish yourself** the number of layers and materials, d_{min}, d_{max}, and total thickness — the simulator does not display them; the *"check my audit"* button is only there to verify your result once you have written it down. Identify layers below the threshold, thicknesses incompatible with the resolution, and any complexity not justified by optical gain. *Point to discuss:* on a quarter-wave stack at 550 nm, the thinnest layer exceeds 55 nm, i.e. more than fifty times the $d_{min} = 1$ nm threshold. Is this guard therefore inactive? Under what circumstances does it become binding again?

💡 Hint

> The layer table gives thicknesses in nm: count the rows, find the smallest and the largest, add them up. “check my audit” is only for checking.

Q29 ★★★

Why three verdicts and not two? At $N = 1000$ and $\Delta d = 2$ nm, record the yield $\hat{p}$ and the p_{low}, p_{high} bounds of its 95% CI. Then choose three values for *Required yield*: one strictly below p_{low}, one between p_{low} and $\hat{p}$, and one strictly above p_{high}. Record and explain the three verdicts. *Why can a design whose point estimate exceeds the requirement remain inconclusive?* What would be needed to lift that verdict — and why does increasing N not guarantee qualification?

💡 Hint

> Read p̂ and the interval at N = 1000, then change only the “Required yield” field. Compare each requirement with both bounds. Course §11.

Q30

★★★

A/B review of the prescription that will actually be manufactured. Compare two states of the *same 18-layer architecture*:

- **Design A — reference:** reload `H L 2H L H L H L 2H L H L H L 2H L H L`, without optimization;
- **Design B — candidate:** start from A exactly, fix layers 1 to 15, and optimize layers 16 to 18 only with BFGS.

Before calculation, write a falsifiable hypothesis: will B improve only the nominal result, or also the robust bound? Qualify A and B with the *same* bands, thresholds, d_{min}, Δd, $N = 1000$, seed, and required yield. Capture A and B in the comparator and require its green valid-paired-comparison indicator.

🧠 **Expected analysis, with no precomputed answer:** compare nominal merit, $\overline{T}_{pass}$, yield with 95% CI, P05(Q_r), P05($\overline{T}_{pass}$), manufacturability, and the paired yield difference with its interval. Identify which indicators move in the same direction and which expose a trade-off.

Give a reasoned decision for B: *adopt*, *reject*, or *extend the study*. A nominal improvement without a statistically defensible robust gain is insufficient. Your conclusion must also confirm that all 18 layers and the first fifteen thicknesses were preserved, and cite at least two limitations of the manufacturing model.

💡 Hint

Capture A then B in the comparator with exactly the same settings: the green light confirms it. Fill in an A | B table, indicator by indicator, before concluding.

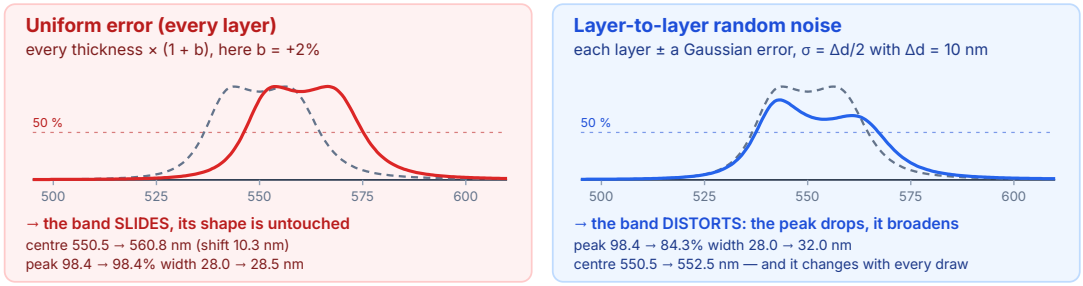
X. Cleanroom reverse engineering & defect diagnosis

⏱ 45 min indicative

🎯 After deposition

Direct entry — prerequisites: have the Q23 nominal spectrum and, if possible, the measured filter spectrum; distinguish spectral shift, shape distortion and transmission loss. Exercises VII and VIII are required for full reverse engineering.

Two process defects, two spectral signatures



A global defect shifts without distorting; a local defect distorts without shifting much.
It is that difference, not the size of the shift, that drives the diagnosis.
Curves computed with the simulator — two-cavity filter BK7 / H L H L 2H L H L H L H L 2H L H L H L / Air, λ_0 = 550 nm, normal incidence, 0.5 nm step; reproc

A **uniform error** multiplies every thickness by the same factor: the stack stays self-similar, so the spectrum translates without distorting. **Independent noise** detunes the layers relative to one another: the centre barely moves but the peak collapses and broadens. Reading this difference is what lets you diagnose a deposition run (Q31). **What matters is not the shift in nanometres but its ratio to the passband width:** the same 5 nm is nearly the whole band on the narrow Q22 filter, and only a tenth of the band on the much wider Fresnel prescription. Always compare the two.

Real cleanroom scenario: the Institut Fresnel deposition team has manufactured your bandpass filter. The spectrophotometer readout shows a **shift of about $8\,\mathrm{nm}$ towards the red** and a drop of the mean transmission measured over 535–565 nm. It is the nominal Q23 prescription that was deposited. Your reference is not the $\lambda_0 = 550\,\mathrm{nm}$ setting, but **the half-maximum centre you measured yourself in Q23**: that is what the discrepancy is measured against. So start by retrieving that reading.

Q31 ★★★

Shifted-window diagnosis. Reproduce three defect scenarios in the simulator and compare their spectral signatures:

- **(a)** a *uniform* scaling of every thickness (multiply them all by the same factor with the **× Scale** button of the layer panel; "↶ Undo" brings back the nominal). Do not change λ_0: index dispersion would distort the factor;
- **(b)** an error on the central $2H$ spacer alone (layer 9);
- **(c)** an error on a single mirror layer — use **layer 5**, so that every pair compares the same thing.

Which one produces a global shift *with the least distortion of the window shape*? What multiplicative factor is needed to reproduce the observed $8\,\mathrm{nm}$? Convert it into a thickness deviation on the thickest layer, then compare this systematic deviation with the Δd severity studied in Q26. Explain why the independent Gaussian model of Q26 cannot, by itself, declare this common bias covered.

Finally, explain why *transmission loss* alone does not identify a unique layer.

💡 **Hint**

Change one factor at a time and memorize each curve to overlay it on the nominal one. For (a), use "× Scale" on all layers; "↶ Undo" brings back the nominal.

Q32
★★★

Metrology and traceability: relate the layer-to-layer random error, set by Δd, to the spread seen across the plotted draws. Derive an optical-thickness tolerance and a resolution requirement from adverse percentiles, not from the nominal result alone. Then translate them into a quartz-monitor recommendation while explicitly listing the missing process data: tooling factor, density, Z-ratio, rate stability, calibration history, and material-specific validation. **No quantified instrument setting may be issued without these data** — state this explicitly in your conclusion.

💡 Hint

> Start from the Q26 percentiles and the shift found in Q31. List what the Monte-Carlo model does not contain before writing your recommendation.

XI. Linear variable filter — research extension

🔬 Not graded · ⏱ 90 min indicative

Direct entry — prerequisites: exercises VI and VII (Bragg mirror, Fabry–Pérot cavity, measuring a full width at half maximum). Questions Q4, Q11 and Q31 are useful. In the simulator, choose exercise **XI**.

💡 **Principle:** on a linear variable filter, the transmitted wavelength changes along the component: $\lambda_c(x) = \lambda_c(0) + g\,x$. This is obtained by varying the deposited thickness with position. On a Helios 800-type magnetron sputtering machine, a mask in front of each target modulates the amount of material received along x: one mask for Nb_2O_5 (H), another for SiO_2 (L). At position x, every H layer is therefore multiplied by a factor $s_H(x)$ and every L layer by $s_L(x)$. **Each position is a stack in its own right**: simulate it with the **× Scale** button of the layer panel (all layers, or a single material), without changing λ_0.

V1
★☆☆

Machine materials. In **+ Material**, create Nb_2O_5 and SiO_2 from the $n(\lambda)$, $k(\lambda)$ tables measured on the Institut Fresnel Helios 800 and published (F. Lemarchand and J. Lumeau, *Optics Continuum*, 2026): download Nb_2O_5 and SiO_2, then import each file choosing the nm unit. Read the indices at $550\,\mathrm{nm}$, compute the contrast $(n_H - n_L)/(n_H + n_L)$ and compare it with the ZnS/YF_3 pair of Q4. In which part of the spectrum does Nb_2O_5 stop being transparent?

💡 Hint

> "+ Material" button, then import the CSV file choosing nm. The table appears: read n and k at 550 nm, then find where k increases.

V2 ★★☆

Reference cavity. On BK7, at $\lambda_0 = 550\,\mathrm{nm}$, enter `(HL)^n 2H (LH)^n`, replacing H and L with the formula codes of your two materials (shown in the editor; for example, if Nb_2O_5 is M2 and SiO_2 M3: `(M2 M3)^3 2M2 (M3 M2)^3`), for $n = 2$, 3 then 4. For each, measure the half-maximum centre, the full width at half maximum and T_{max}. Choose the number of periods giving a resolving power $\lambda/\Delta\lambda$ close to 60 and keep it for what follows.

💡 Hint

> The formula code of each imported material is shown in the editor (M1 is the sample glass provided, your materials follow). Measure the width exactly as in Q22.

V3 ★★☆

The $\lambda_c(s)$ law. Apply the same factor s to every layer ($s_H = s_L = s$) for $s = 0.82$, 1, 1.2, 1.4 and 1.55. Each time, record the centre and the full width at half maximum, then plot λ_c and $\lambda/\Delta\lambda$ against s. Does the centre follow $s \times 550\,\mathrm{nm}$ exactly? On which side of the spectrum is the deviation largest, and which phenomenon already met in Q11 and Q31 explains it? What happens to the width along the component?

💡 Hint

> Use "× Scale" on all layers, starting each time from the s = 1 stack ("↶ Undo"). Build a table: s, centre, width.

V4 ★★★

Profile to deposit. Specification: λ_c from 450 to $850\,\mathrm{nm}$ over $40\,\mathrm{mm}$, i.e. $g = 10\,\mathrm{nm/mm}$, linearity held within $\pm 1\,\mathrm{nm}$. From your V3 curve, determine the factor $s(x)$ that must actually be deposited at five positions or more. Is this thickness profile linear? Quantify the wavelength error made by a mask cut for a thickness that is linear in x.

💡 Hint

> Read your V3 curve backwards: for each target λ, find the s that gives it, by interpolation in Excel.

V5 ★★★

Two masks, two profiles. At $s = 1$, increase by 1% the H layers only, then the L layers only. Record the relative shift of the centre in each case: these are the weights p_H and p_L. Do they add up to 1? Why does the cavity material weigh the most? Deduce the largest admissible gap between $s_H(x)$ and $s_L(x)$ to hold $\pm 1\,\mathrm{nm}$. Finally check on one trial ($s_H = 1.05$, $s_L = 0.95$) what happens to the peak shape: does a mismatch between masks show on the width, on T_{max}, or only on the position?

💡 Hint

> In "× Scale", choose a single material and the factor 1.01. The weight is the relative shift of the centre divided by 1%.

V6 ★★★

Resolution actually measured. A beam of width w illuminates a zone where λ_c varies by $g\,w$. For $w = 0.5$, 1 then $2\,\mathrm{mm}$, export as CSV the spectra of five positions spread over the beam width, average them in Excel and measure the full width at half maximum of the mean spectrum. Compare it with that of V2. Beyond which beam size does adding periods no longer improve the resolution? Propose a rule linking g, w and the intrinsic width.

💡 Hint

> Five positions = five values of s around s = 1, spaced to cover g·w. Export the five spectra and average the T columns in Excel.

V7 ★★☆

Out-of-band blocking. At $s = 1$, widen the display to $350 - 900\,\mathrm{nm}$ and record the interval where $T < 5\%$ on either side of the peak. Compare it with the useful 450–850 nm range: which wavelengths pass through the filter at this position although they should not? Relate the width of the blocked zone to the contrast of V1 and propose two solutions for a real instrument.

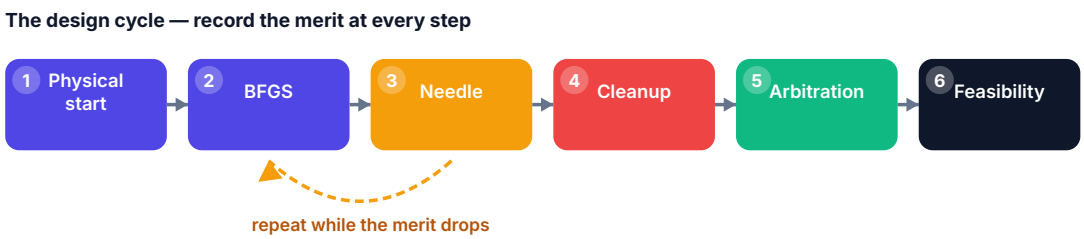
💡 Hint

> Widen λ min and λ max, then find on each side of the peak where T rises above 5% again.

🎨 Open workshop — design your own component

Not graded · self-service

The design cycle, from idea to feasibility dossier



Without step 6, the design is still work in progress, not yet a component.

Steps 2 and 3 alternate: BFGS exploits the existing layers, the needle creates a new one when BFGS saturates. Step 4 is the one people forget, and it decides manufacturability. Step 5 arbitrates between performance and simplicity. Step 6 is the only one that turns a calculation into a component.

> The ten main exercises imposed specifications on you. **Here, you write them.** The simulator accepts any spectral target: it becomes a reusable design tool well beyond this lab — for a project, an internship, or simply to test an idea.

🚀 Declare the target — 30 seconds

1. Press `D` — or click 🎛 **Free** in the header. A six-step guide shows the next action; fine settings (**Clean up**, **Analyze the needle**, minimum thickness, required yield) are folded under **More options** ▾.
2. Click **+ Segment**, then translate each requirement into one row of the table.
3. Enter λ min, λ max, T/R/A quantity, angle, polarization, target and weight.
4. Save it (↓): this is your specification, and it must not move during design.

⚙️ What the merit computes

The score is a **weighted RMSE** between your computed response and your target, over the declared segments only:

$$\mathrm{RMSE} = \sqrt{\frac{\sum_i w_i (X_i - C_i)^2}{\sum_i w_i}}$$

Anything you do not cover is **"don't care"** and does not enter the score. The weight w_i lets you state that one band matters more than another. This is exactly the structure of an industrial specification.

🛠️ The designer's method — this is a craft, not a click

Declaring a target and pressing "Optimize" almost never produces a good component. Real design is an **iterative cycle** that you must run yourself, and it is not finished until the feasibility study.

1. **Choose a starting solution — never at random.**
   The optimizer is *local*: it descends into whichever valley you drop it in. Start from a structure that physics justifies — a quarter-wave for antireflection, `(HL)^n` for a mirror, `(HL)^n 2H (LH)^n` for a bandpass — centred on the right wavelength. A good initialization beats any algorithm.

2. **Refine the thicknesses with Quasi-Newton (BFGS).**
   Run the **Quasi-Newton (BFGS)** optimizer with key `O`. It adjusts thicknesses *at constant structure*: it will never create a layer, it only exploits the ones you give it. Record the merit before and after.

3. **Enrich the structure: needle → optimize cycles.**
   When BFGS stops progressing, the *structure* is the limit, not the thicknesses. Run **+ Useful layer (needle)**, or **Analyze the needle** to see the diagnostic (key `N`): the $\pi(z)$ diagnostic tells you where to insert an ultra-thin layer of the other material. Insert, then **immediately re-optimize with BFGS**. Repeat.

   When to stop? When a full cycle no longer lowers the merit significantly, or when the needle finds no positive maximum. Keep a log: layer count and merit at every cycle. The curve "merit versus number of layers" *is* a design result.

4. **Clean the stack — the step everyone forgets.**
   After a few cycles your stack contains 0.1 nm slices and adjacent layers of the same material. **None of them is depositable.** Two tools:
   - **Clean up button** (panel *2. Layer stack*, under *More options* ▾): set a threshold; it removes every thinner layer *and merges* adjacent layers of the same material. It then reports how many layers you saved and how much merit you lost.
   - **Row-by-row deletion:** the *Action* column of the layer table, to remove one specific layer you judge useless.

Then re-optimize: the remaining thicknesses must absorb what was removed. You will often be surprised — a 7-layer stack can shrink to 2 with no measurable loss. A layer that costs nothing to delete was doing nothing.

5 **Arbitrate: performance versus complexity.**
Every added layer costs machine time, one more error source and a scrap risk. Compare your successive designs: how many merit points does the 8th layer buy? If the answer is "almost none", the engineering decision is to drop it. **The best design is not the highest performing one: it is the simplest one that meets the specification.**

6 **Close with the feasibility study — otherwise the work is not done.**
A design exists only if it can be deposited. Apply the lab protocol to *your* component:

- **Nominal audit**: number of layers and materials, d_{min}, d_{max}, total thickness. A layer below the deposition threshold disqualifies the design.
- **Statistical qualification**: *Assess robustness*, $N = 1000$. Report yield, 95% CI, adverse percentile and manufacturability rate.
- **Sensitivity to Δd**: determine the deposition accuracy to require from the process to hold the target yield.
- **Decision**: *qualified*, *inconclusive* (study to be extended) or *not qualified*, with your assumptions and at least two model limitations.

An "optimized" design without a feasibility study is still work in progress: feasibility is what turns it into a component.

> **The cycle in one line:** `physical start → BFGS → (needle → BFGS)ⁿ → cleanup → BFGS → arbitration → feasibility`. Record the merit at every step: that is your design trace, and it is what makes your component defensible.

⚠️ **Three traps this lab taught you to avoid — they apply here too:**

- **Do not specify what physics forbids.** A T = 100% target over 400–1000 nm on an absorbing substrate is unreachable: the optimizer will converge to a poor compromise without warning you.
- **Leave don't-care zones.** A sharp transition needs room: if you specify 100% at 550 nm and 0% at 555 nm, no finite stack will achieve it.
- **A good nominal score is not enough on its own.** Always run the statistical qualification before declaring a design good.

🎯 Six challenges, from simplest to most demanding

★☆☆ **Hot mirror.** $R \geq 95\%$ *at every point* of $700 - 850$ nm, and $\overline{T} \geq 85\%$ over $420 - 650$ nm: it reflects the near infrared and transmits the visible — the opposite of a cold mirror. How many layers do you need? *Hint: a quarter-wave stack centred at 775 nm gets there; find the minimum count.*

★☆☆ **Single-colour antireflection.** Aim for $\overline{R} \leq 0.2\%$ over $530 - 570$ nm only, with ZnS/YF_3 on BK7. *A single YF_3 layer cannot get there: compare $n_L = 1.469$ with the ideal index $\sqrt{n_{BK7}} = 1.232$ from Q10. Show that a second material removes this limit (think of Q13), then compare the width of your $R < 0.2\%$ zone with that of the broadband design of Q17: that is the price of bandwidth.*

★★☆ **Notch filter.** Try to block $T \leq 1\%$ over $525 - 545$ nm while keeping $\overline{T} \geq 80\%$ everywhere else in the visible. *With a quarter-wave stack $(HL)^n H$, measure the width obtained below 10% transmission: once the rejection reaches 1%, it exceeds 150 nm whatever n. Relate that limit to the index contrast of Q4. Then try three-quarter-wave layers, $(3H\,3L)^n\,3H$: what happens to the width, and why? For a 20 nm notch, which levers remain — higher order, another material pair?*

★★☆ **50/50 beamsplitter.** $R = T = 50\%$ over $450 - 650$ nm, target RMSE ≤ 1 point. *A single ZnS layer tops out at about one third reflectance: several layers must be combined to reach 50%, then*

the response must be flattened over 200 nm. Plot the best RMSE you obtain against the number of layers, from 1 to 7: beyond how many layers does the gain become marginal? Why does one more layer not always help, and what does this curve reveal about the local nature of BFGS?

★★★ **Your own, qualified.** Write a full specification, meet it, then qualify it ($N = 1000$). Submit the Q27–Q30 decision sheet on *your* component.

★★★ **The duel.** Have another team design a target, swap JSON files, and qualify each other's design. Defend or refute their conclusion.

> **Keep your work:** the ↓ and ↑ buttons in the editor export and re-import your target as JSON. A saved specification can be reused, passed to a classmate, or attached to a report as supporting evidence.

📚 Bibliography & Recommended Resources

- **Course Document:** Theory of Optical Thin Films.
- **The Optical Coating Bible:** H.A. Macleod, *"Thin-Film Optical Filters"* (CRC Press).
- **Global Dispersion Database:** refractiveindex.info.
- **Theoretical Foundations:** Max Born & Emil Wolf, *"Principles of Optics"*.

BUT 3 FI MCPC Path — L2 · L3 · M1 · M2 students — Institut Fresnel 2026

Modular optical thin-film simulation lab path

Online version and interactive simulator: tpcouchesminces.vercel.app